



U.S. Environmental Protection Agency | Region 2

National Pollutant Discharge Elimination System

Reasonable Potential Analysis Report

Permit PR0000698

PUERTO RICO ELECTRIC PWR AUTH

Monitoring period 2021-10-06 to 2026-10-06

Application forms 2C, 2F

Report generated October 06, 2026

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1 Summary

Permit: PR0000698

Facility: PUERTO RICO ELECTRIC PWR AUTH

Date range: 2021-10-06 to 2026-10-06

Water class(es): class SB waters

Selected forms: 2C, 2F

Dilution ratio: 1

Confidence level: 0.95

Target percentile: 0.95

Hardness setting: Range

1.1 Included Pollutants

The following table shows the available data for each pollutant included in this analysis. The *Basis for Inclusion* column records why each pollutant is present: required by a selected NPDES form (2C, 2F); carried in because prior DMR data exists even though no selected form requires it (*Existing DMR*); or present only because the user supplied the monitoring data (*Manually Added*). A † marker flags parameters whose record includes monitoring data manually added by the user on top of the base DMR record. For a complete list of pollutants that had no associated data, refer to Section 5.

Pollutant	Basis for Inclusion	Ref Param
Copper	Form 2C	01042
Cyanide	Form 2C	51173
Dissolved oxygen	Existing DMR	00300
Enterococci	Existing DMR	61211
Lead	Form 2C	01051
Mercury	Form 2C	71900
Nickel	Form 2C	01067
Polychlorinated biphenyls [PCBs]	Existing DMR	39516
Selenium	Form 2C	01147
Sulfate, total [as SO ₄]	Existing DMR	00945
Temperature	Existing DMR	00010 †
Turbidity	Existing DMR	00070
Zinc	Form 2C	01092
pH (maximum)	Forms 2C & 2F	00400
pH (minimum)	Forms 2C & 2F	00400

† WQS criterion manually entered by user; see Manual WQS Overrides section.



2 RP Findings

RP column: **YES** indicates the assessed value exceeds the criterion. **DEPENDS** indicates an exceedance was not observed but the tool cannot conclude no reasonable potential — see the note on the pollutant's section. **NO** indicates no exceedance.

The Basis column distinguishes a *Calculated* receiving water concentration (TSD multiplying-factor projection, dilution ratio applied) from an *Observed* maximum or minimum compared directly to the criterion with no projection and no dilution. Values rounded to 2 decimals.

WQS ID	Pollutant	Outfall	Water Class	WQS Criterion	Value	Basis	RP
79235	Copper	External Outfall 002	class SB waters	3.73	128.18	Calculated	YES
79235	Copper	External Outfall 003	class SB waters	3.73	54.90	Calculated	YES
79232	Cyanide	External Outfall 002	class SB waters	1.00	2.28	Calculated	YES
79590	Dissolved Oxygen	External Outfall 001	class SB waters	5.00	4.10	Observed min	YES
79590	Dissolved Oxygen	External Outfall 002	class SB waters	5.00	3.30	Observed min	YES
79590	Dissolved Oxygen	External Outfall 003	class SB waters	5.00	3.40	Observed min	YES
79594	Enterococci (90th percentile)	External Outfall 003	class SB waters	130.00	10000.00	Observed max	YES
79593	Enterococci (geometric mean)	External Outfall 003	class SB waters	35.00	69000.00	Observed max	YES
79244	Mercury	External Outfall 002	class SB waters	0.05	0.08	Calculated	YES
79247	Nickel	External Outfall 002	class SB waters	8.28	65.05	Calculated	YES
79247	Nickel	External Outfall 003	class SB waters	8.28	47.47	Calculated	YES
MANUAL- 00010	Temperature, water deg. centigrade	External Outfall 002	class SB waters	30.00	36.10	Observed max	YES
MANUAL- 00010	Temperature, water deg. centigrade	External Outfall 003	class SB waters	30.00	38.90	Observed max	YES
79596	Turbidity	External Outfall 003	class SB waters	10.00	13.20	Observed max	YES
79263	Zinc	External Outfall 001	class SB waters	85.62	105.58	Calculated	YES
79263	Zinc	External Outfall 002	class SB waters	85.62	390.94	Calculated	YES
79263	Zinc	External Outfall 003	class SB waters	85.62	163.36	Calculated	YES
79595	pH (maximum)	External Outfall 003	class SB waters	8.50	9.30	Observed max	YES
79595	pH (maximum)	Internal Outfall 601	class SB waters	8.50	9.00	Observed max	YES
79595	pH (minimum)	External Outfall 002	class SB waters	7.30	5.00	Observed min	YES
79595	pH (minimum)	External Outfall 003	class SB waters	7.30	4.50	Observed min	YES



(continued)

WQS ID	Pollutant	Outfall	Water Class	WQS Criterion	Value	Basis	RP
79595	pH (minimum)	Internal Outfall 601	class SB waters	7.30	6.50	Observed min	YES
79595	pH (minimum)	Internal Outfall 603	class SB waters	7.30	6.30	Observed min	YES
79595	pH (minimum)	Internal Outfall 604	class SB waters	7.30	6.70	Observed min	YES
MANUAL- 00010	Temperature, water deg. centigrade	External Outfall 001	class SB waters	30.00	-8.39	Observed max	DEPENDS
79252	Lead	External Outfall 002	class SB waters	8.52	5.34	Calculated	NO
79252	Lead	External Outfall 003	class SB waters	8.52	2.66	Calculated	NO
79401	Polychlorinated Biphenyls (PCBs)	Internal Outfall 601	class SB waters	0.00	0.00	Calculated	NO
79401	Polychlorinated Biphenyls (PCBs)	Internal Outfall 602	class SB waters	0.00	0.00	Calculated	NO
79401	Polychlorinated Biphenyls (PCBs)	Internal Outfall 603	class SB waters	0.00	0.00	Calculated	NO
79401	Polychlorinated Biphenyls (PCBs)	Internal Outfall 604	class SB waters	0.00	0.00	Calculated	NO
79401	Polychlorinated Biphenyls (PCBs)	Internal Outfall 605	class SB waters	0.00	0.00	Calculated	NO
79255	Selenium	External Outfall 002	class SB waters	71.14	50.40	Calculated	NO
79597	Sulfate, total [as SO ₄]	External Outfall 001	class SB waters	2800.00	2748.08	Calculated	NO
79596	Turbidity	External Outfall 001	class SB waters	10.00	7.72	Observed max	NO
79595	pH (maximum)	External Outfall 001	class SB waters	8.50	8.30	Observed max	NO
79595	pH (maximum)	External Outfall 002	class SB waters	8.50	8.50	Observed max	NO
79595	pH (maximum)	Internal Outfall 602	class SB waters	8.50	8.50	Observed max	NO
79595	pH (maximum)	Internal Outfall 603	class SB waters	8.50	8.20	Observed max	NO
79595	pH (maximum)	Internal Outfall 604	class SB waters	8.50	8.00	Observed max	NO
79595	pH (maximum)	Internal Outfall 605	class SB waters	8.50	8.40	Observed max	NO
79595	pH (minimum)	External Outfall 001	class SB waters	7.30	7.60	Observed min	NO
79595	pH (minimum)	Internal Outfall 602	class SB waters	7.30	8.00	Observed min	NO
79595	pH (minimum)	Internal Outfall 605	class SB waters	7.30	7.70	Observed min	NO



3 Non-Concentration Based Measures

Parameters in this section are assessed by direct comparison: the observed maximum (for a ceiling criterion) or minimum (for a floor) is compared to the criterion as reported. No multiplying factor is applied and the dilution ratio is not used.

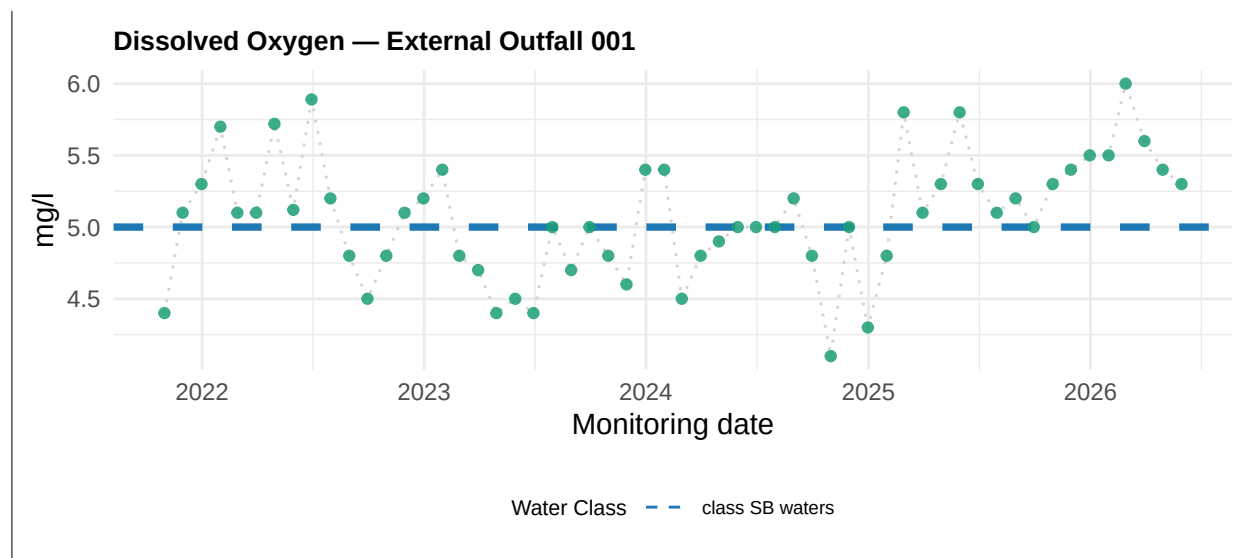
3.1 Dissolved Oxygen

Note — Dissolved oxygen is evaluated by direct comparison of the observed minimum to the criterion; no RWC is calculated. A defensible DO analysis requires the Streeter-Phelps equation, whose inputs (reaeration and deoxygenation rates, travel time, upstream deficit) are not available in ICIS-NPDES. Where no excursion is observed the determination is DEPENDS, not NO: the permit writer should determine whether additional data must be collected to apply Streeter-Phelps.

This parameter is assessed by direct comparison. The observed minimum is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.

3.1.1 External Outfall 001

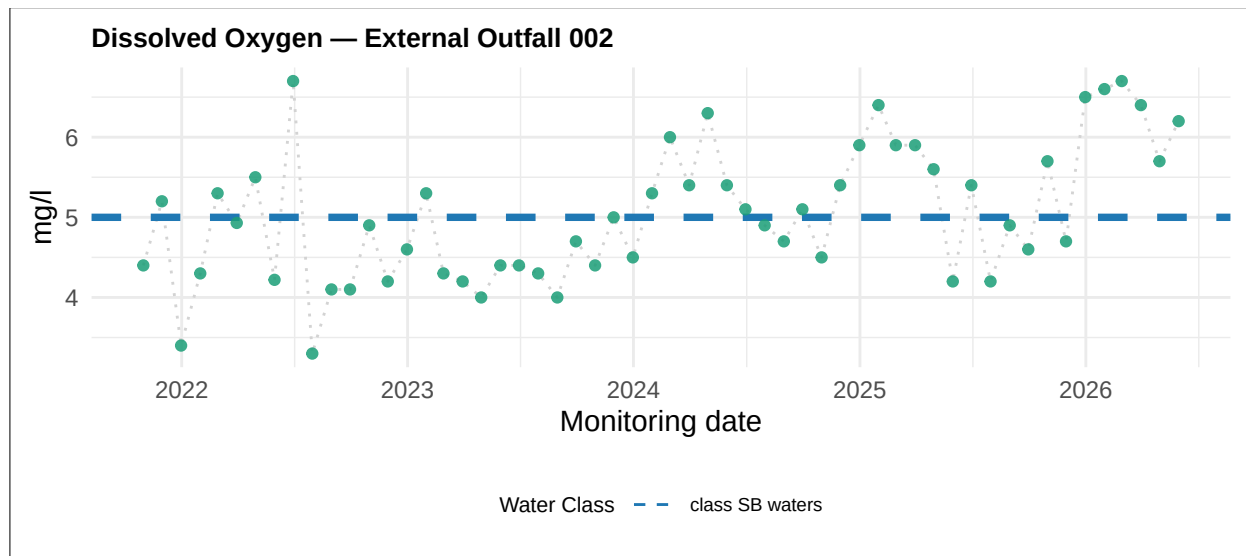
WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Min	RP
79590	class SB waters	56	No	5	4.1	YES





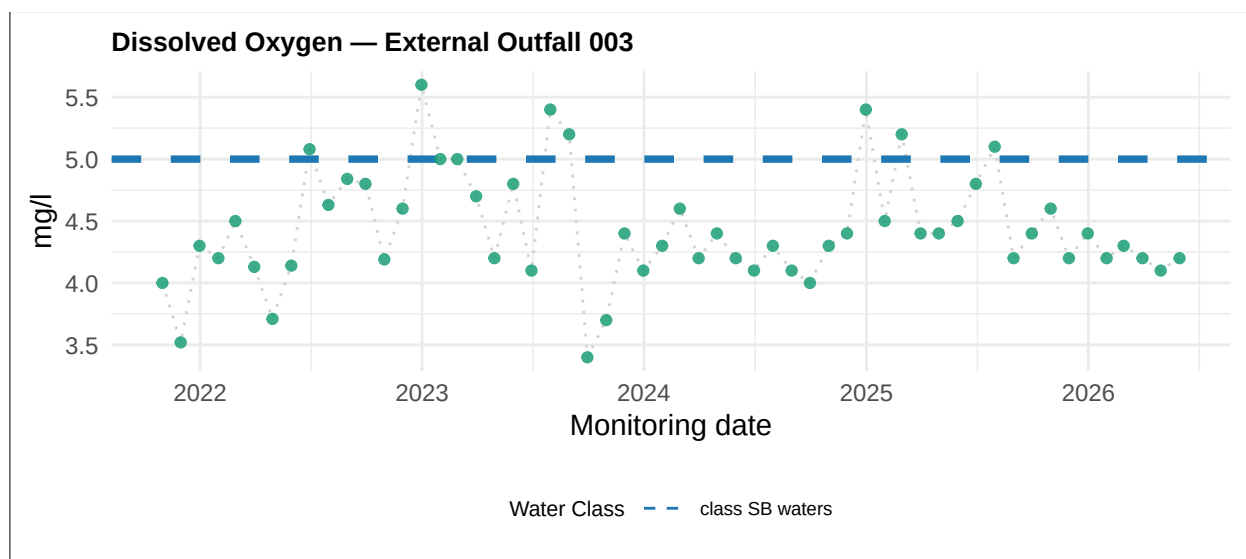
3.1.2 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Min	RP
79590	class SB waters	56	No	5	3.3	YES



3.1.3 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Min	RP
79590	class SB waters	56	No	5	3.4	YES





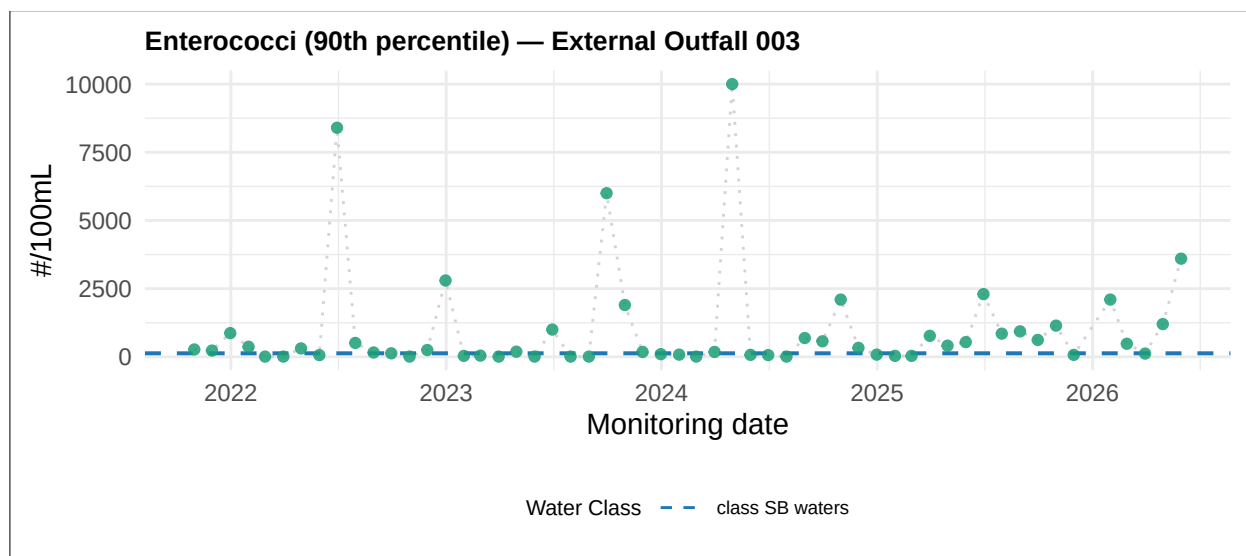
3.2 Enterococci (90th percentile)

Note — Enterococci is not evaluated by a calculated RWC. Both PRWQS criteria are frequency- and interval-based — a geometric mean over any 90-day interval (35/100 mL) and the 90th percentile of samples over that same interval (130/100 mL) — and neither is a maximum concentration to which a multiplying-factor projection can be meaningfully applied. Reported values are compared directly to the criteria. DMR geometric means are commonly 30-day or monthly rather than 90-day, and dilution is not applied. The permit writer should evaluate the geometric mean and the 90th percentile over the applicable 90-day intervals independently before concluding on reasonable potential.

This parameter is assessed by direct comparison. The observed maximum is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.

3.2.1 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Max	RP
79594	class SB waters	55	No	130	10000	YES





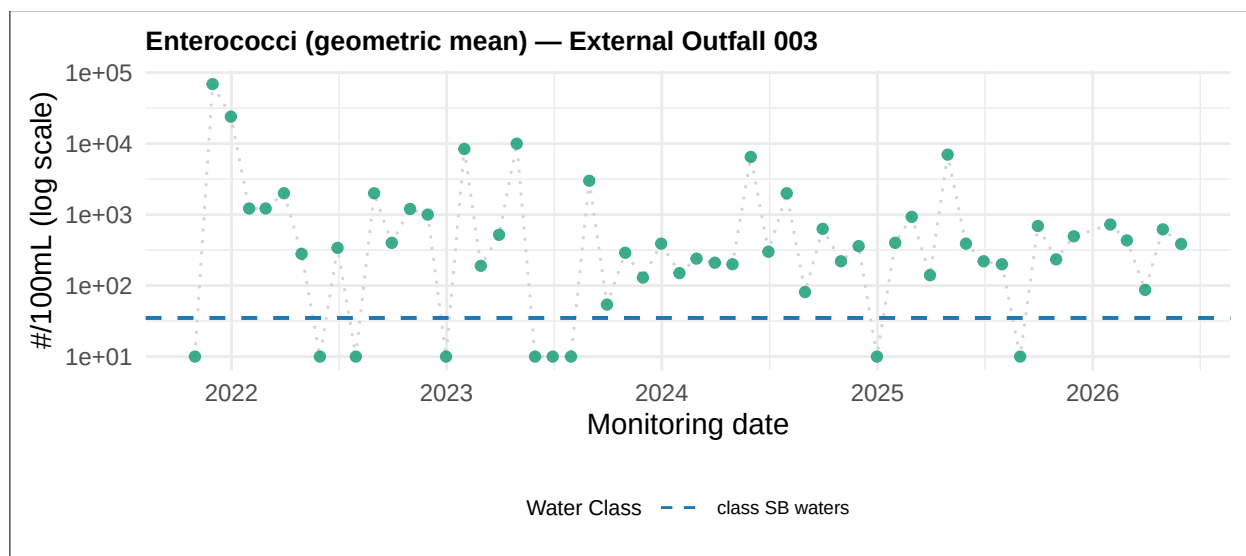
3.3 Enterococci (geometric mean)

Note — Enterococci is not evaluated by a calculated RWC. Both PRWQS criteria are frequency- and interval-based — a geometric mean over any 90-day interval (35/100 mL) and the 90th percentile of samples over that same interval (130/100 mL) — and neither is a maximum concentration to which a multiplying-factor projection can be meaningfully applied. Reported values are compared directly to the criteria. DMR geometric means are commonly 30-day or monthly rather than 90-day, and dilution is not applied. The permit writer should evaluate the geometric mean and the 90th percentile over the applicable 90-day intervals independently before concluding on reasonable potential.

This parameter is assessed by direct comparison. The observed maximum is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.

3.3.1 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Max	RP
79593	class SB waters	55	No	35	69000	YES



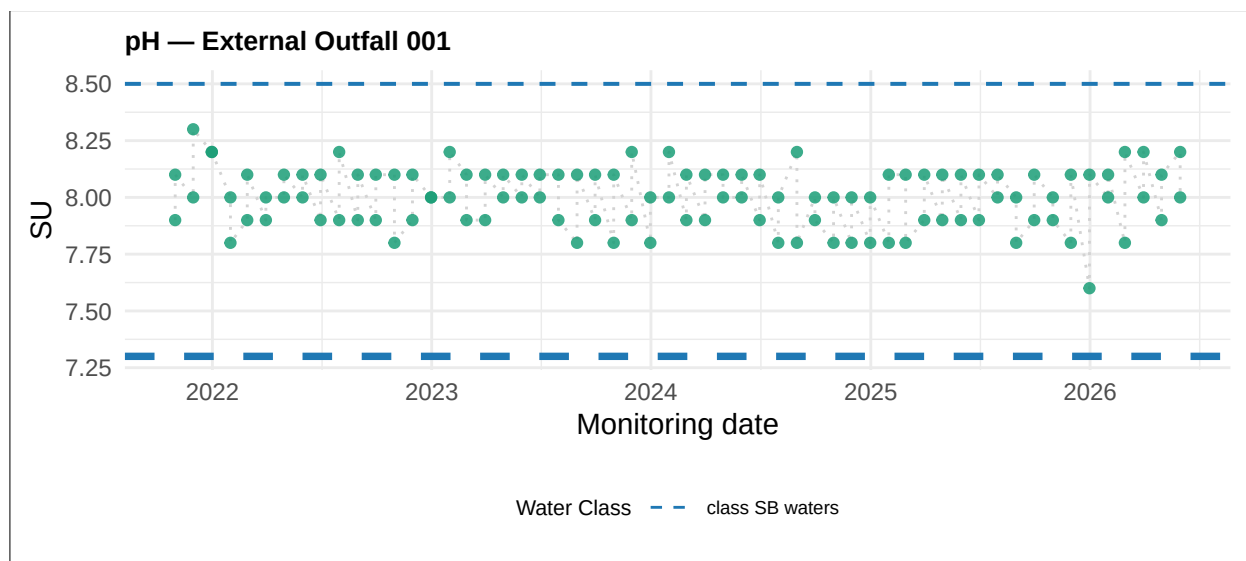


3.4 pH

This parameter is assessed by direct comparison. The observed maximum is compared to the upper criterion and the observed minimum to the lower criterion as reported: no multiplying factor is applied and the dilution ratio is not used.

3.4.1 External Outfall 001

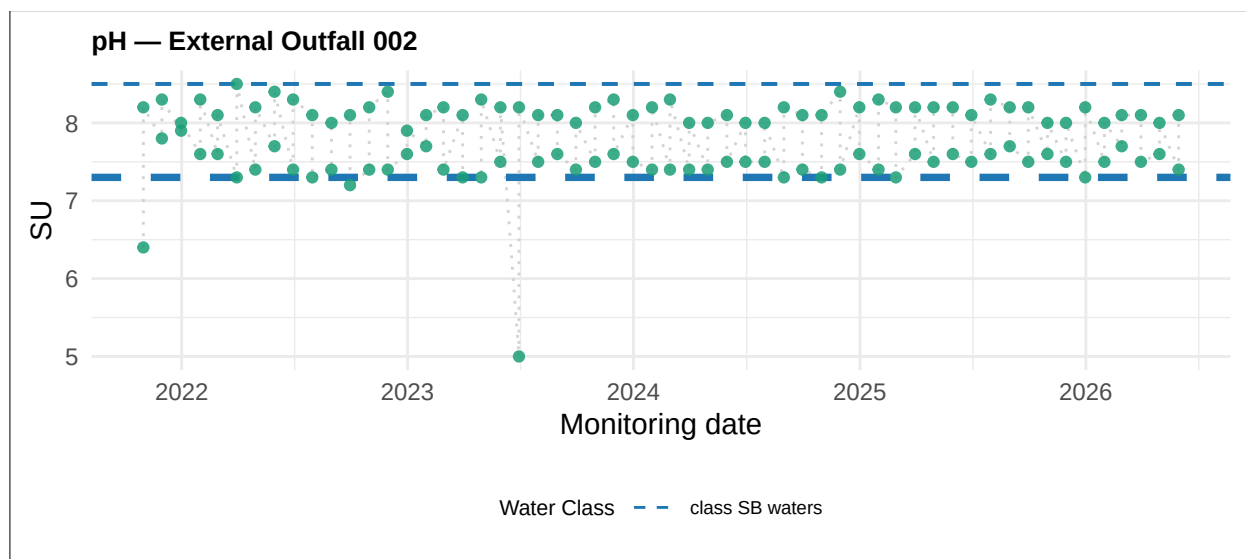
WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	56	No	7.3	7.6	NO
79595	class SB waters	Max	56	No	8.5	8.3	NO





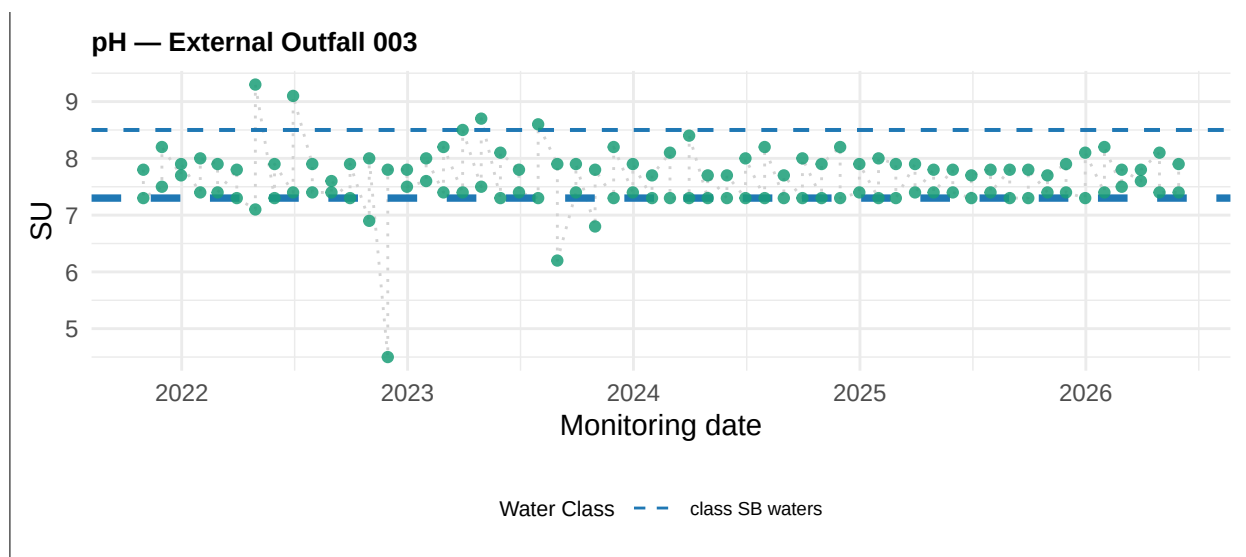
3.4.2 External Outfall 002

WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	56	No	7.3	5.0	YES
79595	class SB waters	Max	56	No	8.5	8.5	NO



3.4.3 External Outfall 003

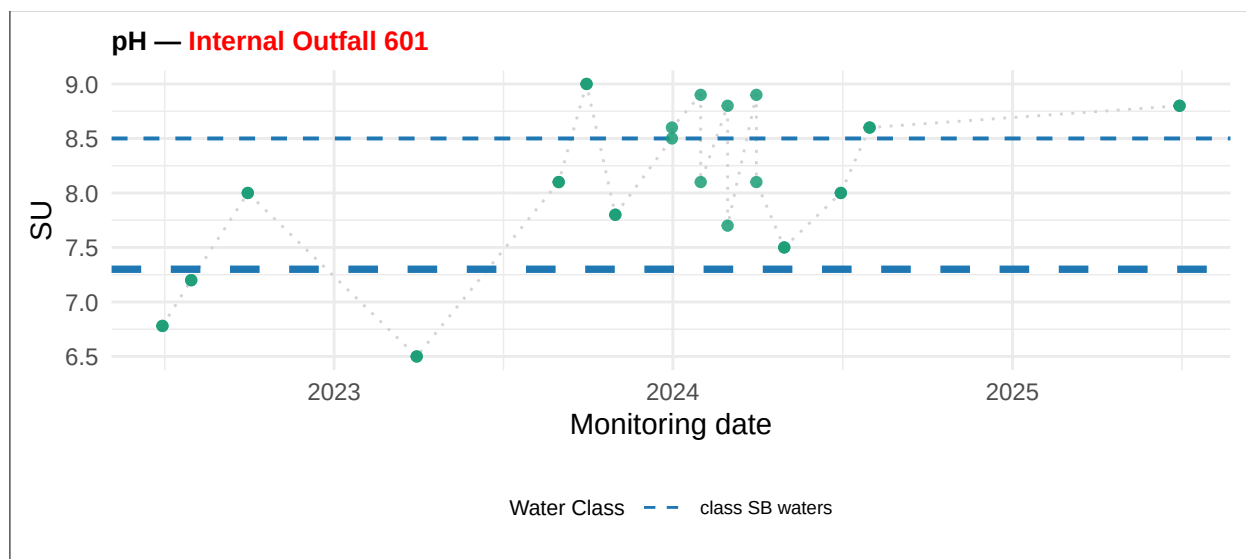
WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	56	No	7.3	4.5	YES
79595	class SB waters	Max	56	No	8.5	9.3	YES





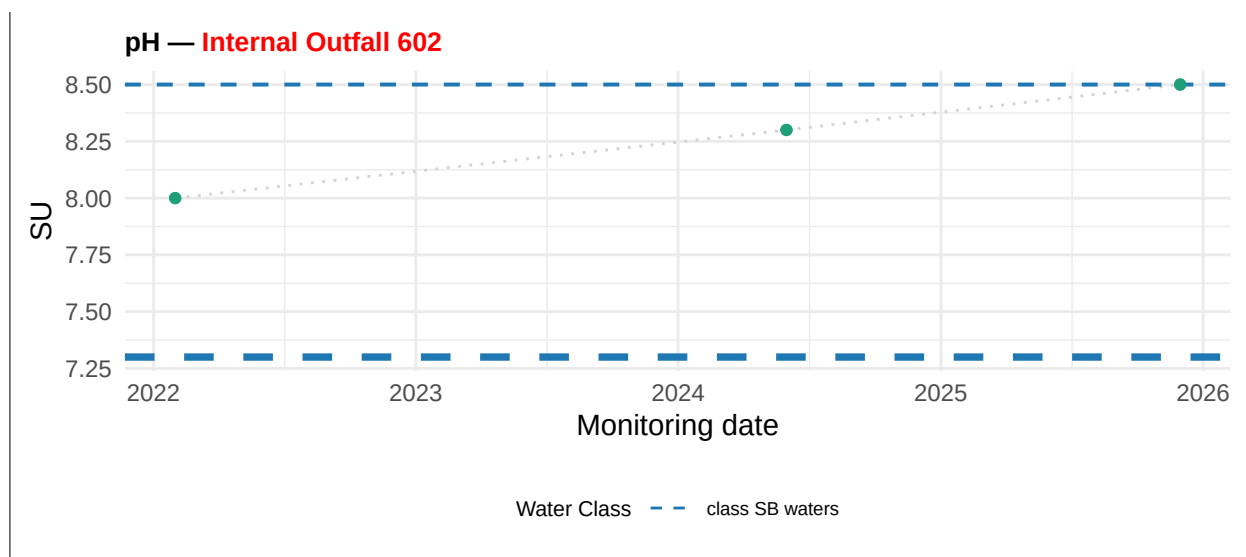
3.4.4 Internal Outfall 601

WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	15	No	7.3	6.5	YES
79595	class SB waters	Max	15	No	8.5	9.0	YES



3.4.5 Internal Outfall 602

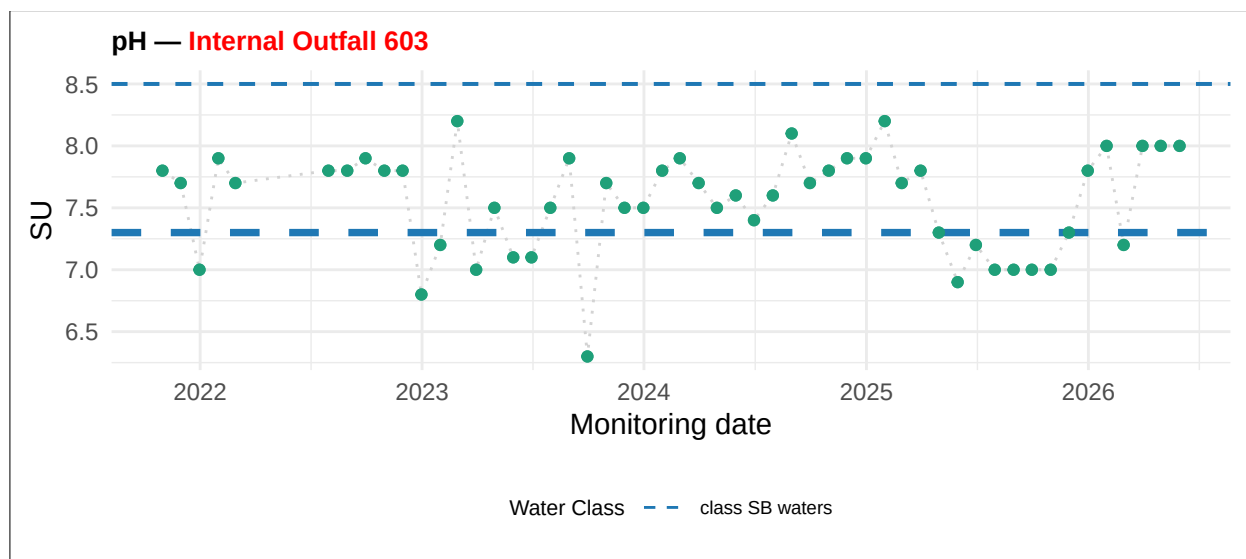
WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	3	No	7.3	8.0	NO
79595	class SB waters	Max	3	No	8.5	8.5	NO





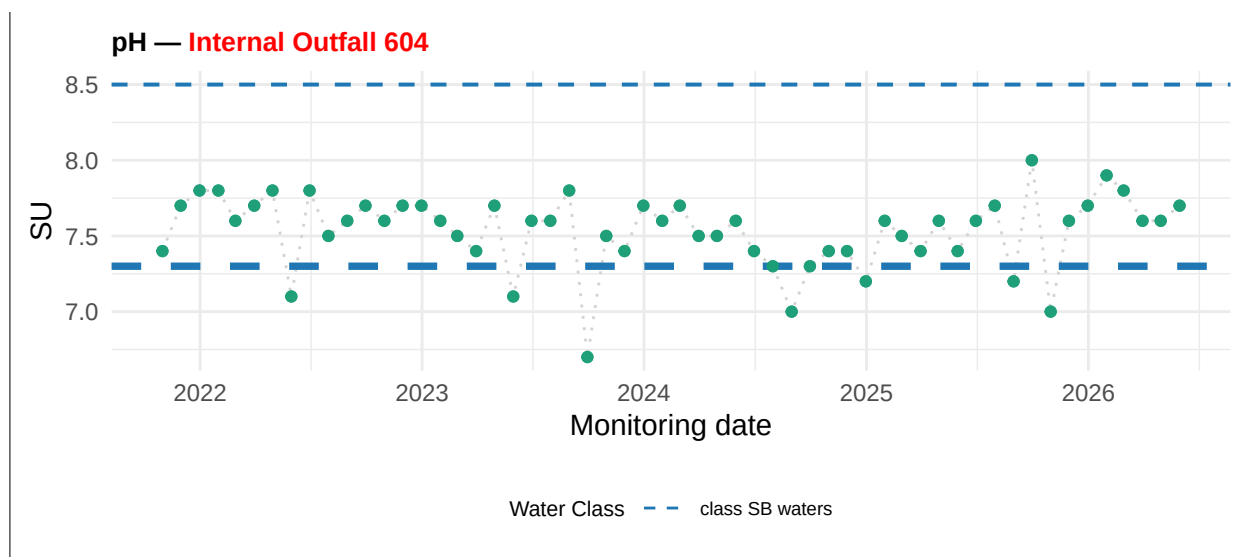
3.4.6 Internal Outfall 603

WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	52	No	7.3	6.3	YES
79595	class SB waters	Max	52	No	8.5	8.2	NO



3.4.7 Internal Outfall 604

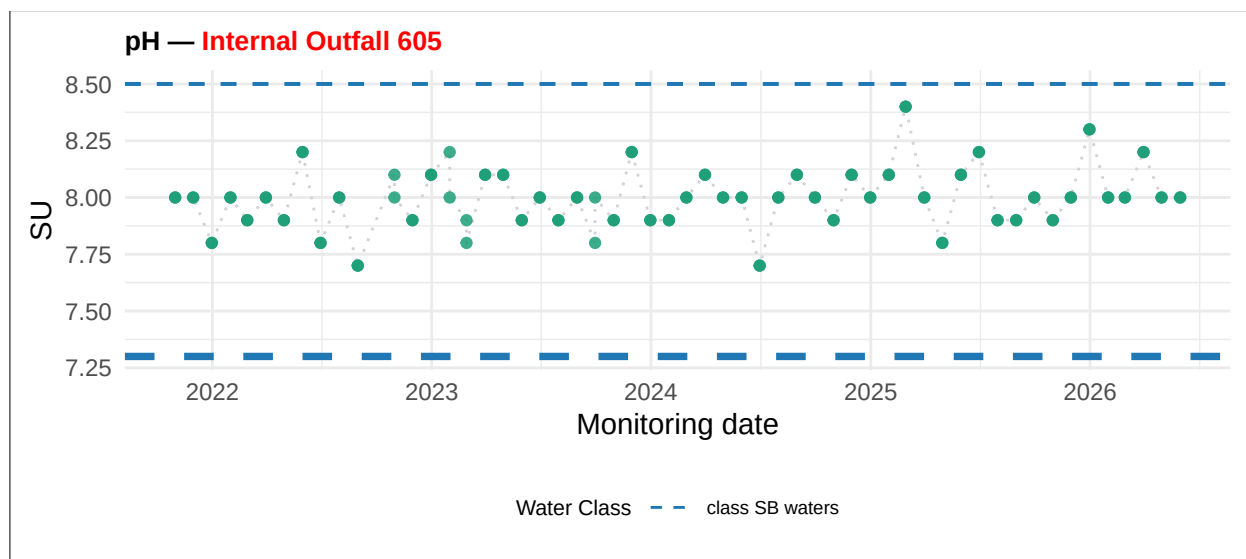
WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	56	No	7.3	6.7	YES
79595	class SB waters	Max	56	No	8.5	8.0	NO





3.4.8 Internal Outfall 605

WQS ID	Water Class	Bound	Samples	Units Converted	WQS Criterion	Observed	RP
79595	class SB waters	Min	55	No	7.3	7.7	NO
79595	class SB waters	Max	55	No	8.5	8.4	NO





3.5 Temperature, water deg. centigrade

Note — The PRWQS temperature criterion applies in the receiving water — a discharge shall not cause the temperature at any site to exceed 30 C — not at the point of discharge. An end-of-pipe comparison is therefore screening only. Where no excursion is observed the determination is **DEPENDS**: the permit writer should evaluate in-stream temperature and the applicable mixing zone before concluding on reasonable potential.

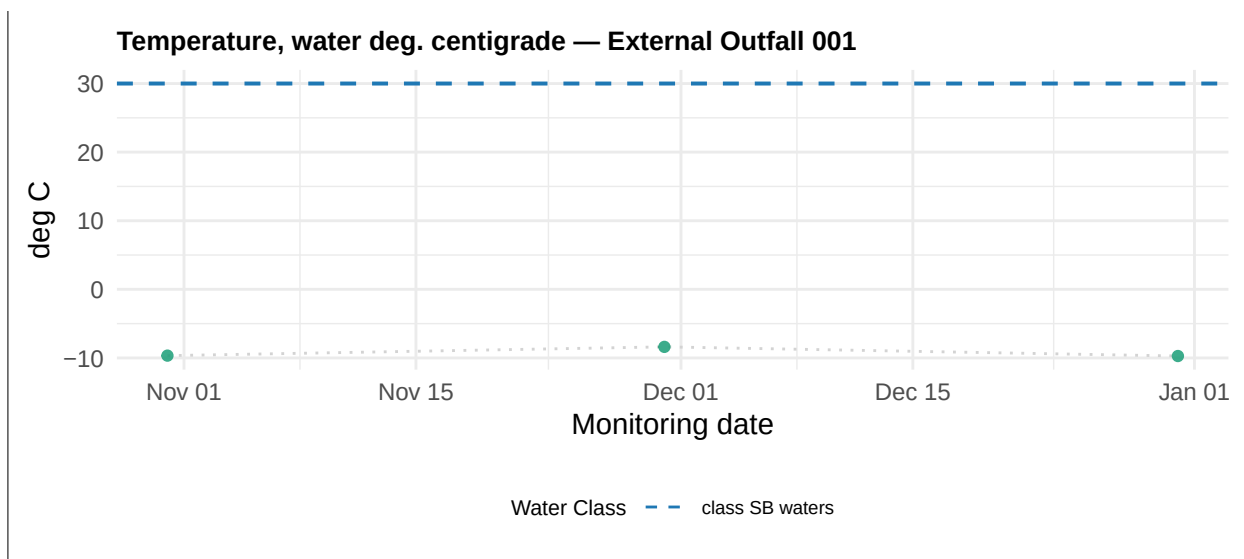
Manual WQS Applied — Criterion value manually entered by user. See *Manual WQS Overrides* section for full details.

This parameter is assessed by direct comparison. The observed maximum is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.

3.5.1 External Outfall 001

Check data — 3 of 3 temperature values at this outfall are below 0 °C. They were reported under parameter 00011 (°F) and converted to °C; values below freezing suggest they were actually reported in °C. Verify against the DMR before relying on this result.

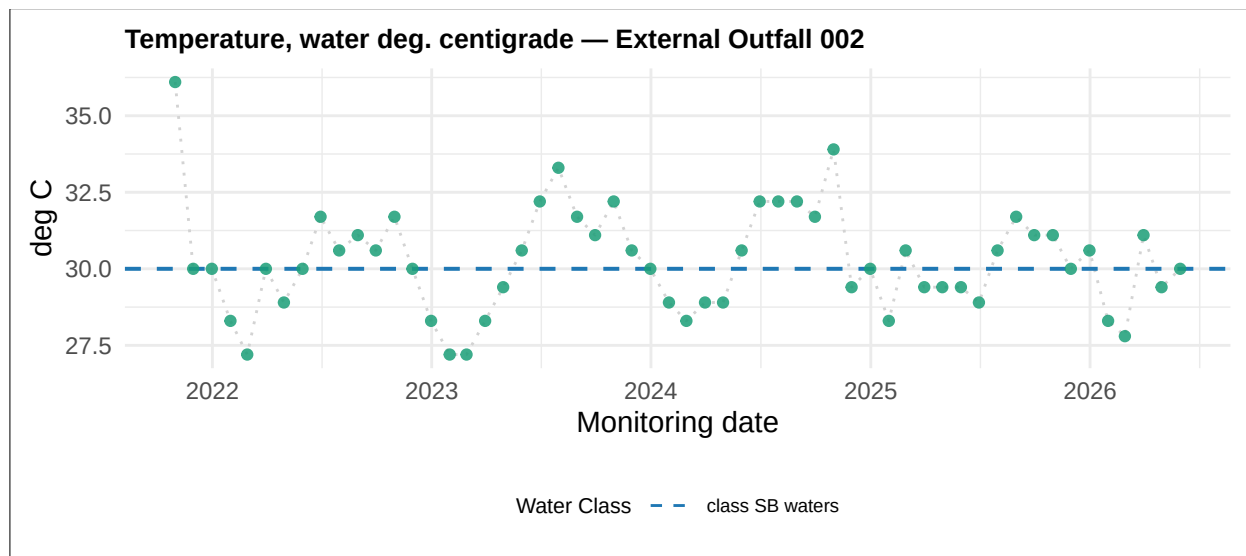
WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Max	RP
MANUAL-00010	class SB waters	3	Yes	30	-8.3889	DEPENDS





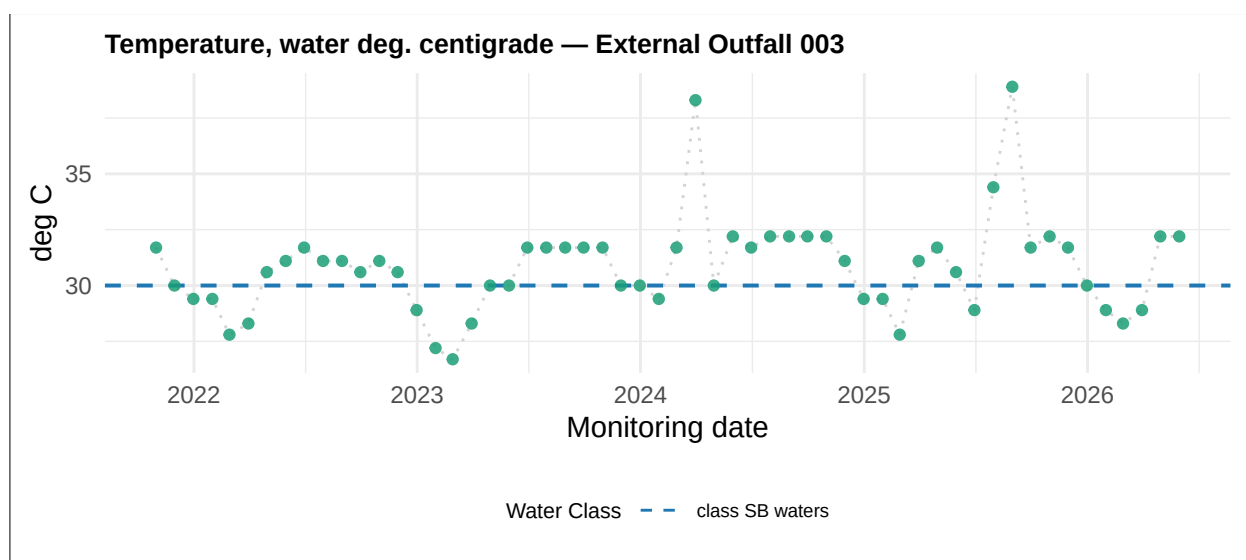
3.5.2 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Max	RP
MANUAL-00010	class SB waters	56	No	30	36.1	YES



3.5.3 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Max	RP
MANUAL-00010	class SB waters	56	No	30	38.9	YES



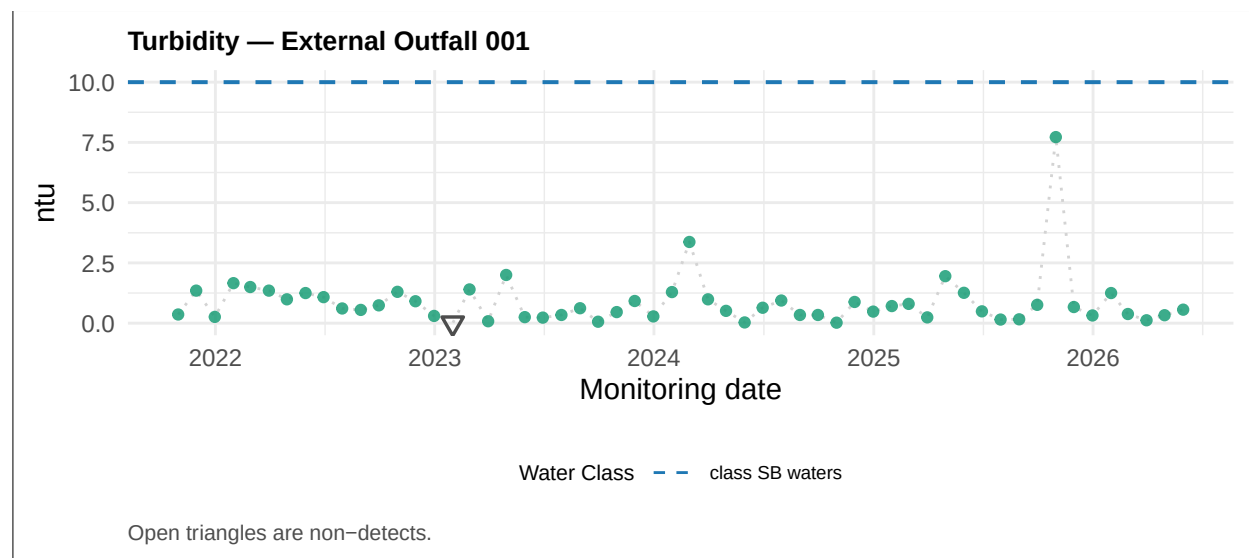


3.6 Turbidity

This parameter is assessed by direct comparison. The observed maximum is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.

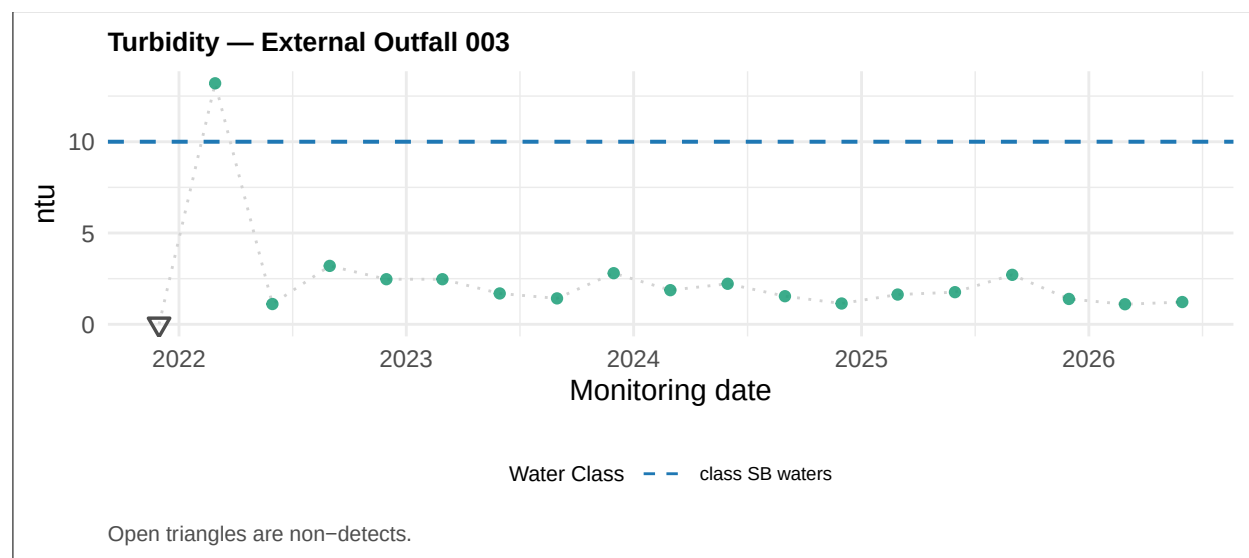
3.6.1 External Outfall 001

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Max	RP
79596	class SB waters	56	No	10	7.72	NO



3.6.2 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	Observed Max	RP
79596	class SB waters	19	No	10	13.2	YES





4 Concentration Based Measures

4.1 RP Equations

The following equations estimate the Receiving Water Concentration (RWC) for parameters assessed by projection. Parameters assessed by direct comparison are reported in the Non-Concentration Based Measures section.

$$\sigma_g = \sqrt{\ln(CV^2 + 1)}$$

$$p = (1 - CL)^{1/n}$$

$$z_p = \Phi^{-1}(p), z_{\text{target}} = \Phi^{-1}(\text{target})$$

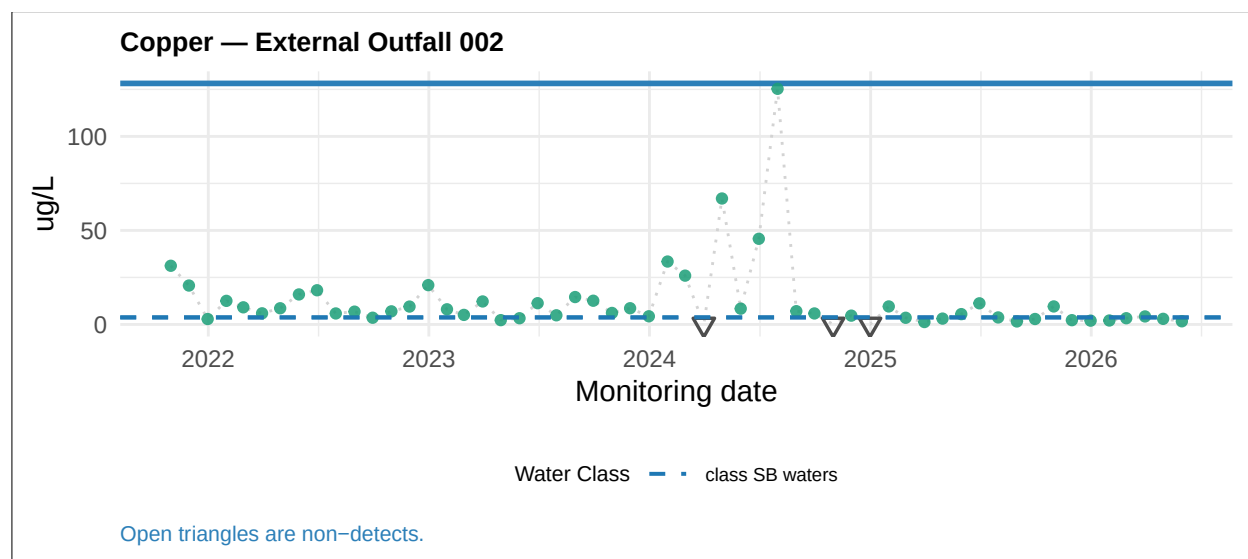
$$MF = \exp((z_{\text{target}} - z_p) \sigma_g)$$

$$RWC = \frac{\max(x) \times MF}{DR}$$

4.2 Copper

4.2.1 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79235	class SB waters	56	No	3.73	128	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(1.6^2 + 1)} = 1.1$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

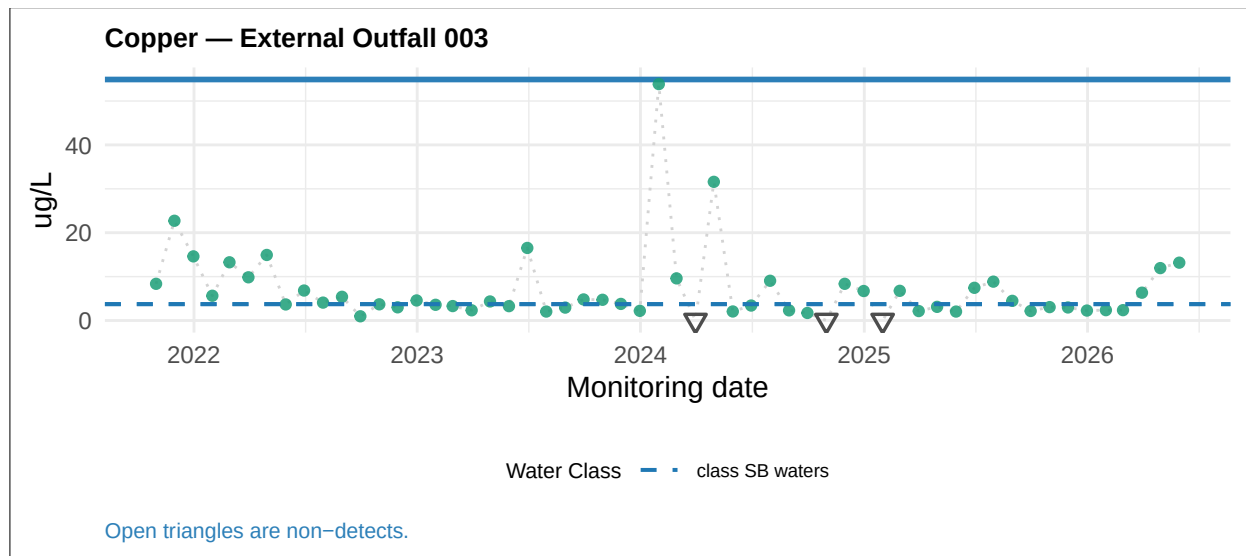
$$MF = \exp((1.6 - 1.6) \times 1.1) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{130 \times 1}{1} = 130$$



4.2.2 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79235	class SB waters	56	No	3.73	54.9	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(1.2^2 + 1)} = 0.97$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.6) \times 0.97) = 1$$

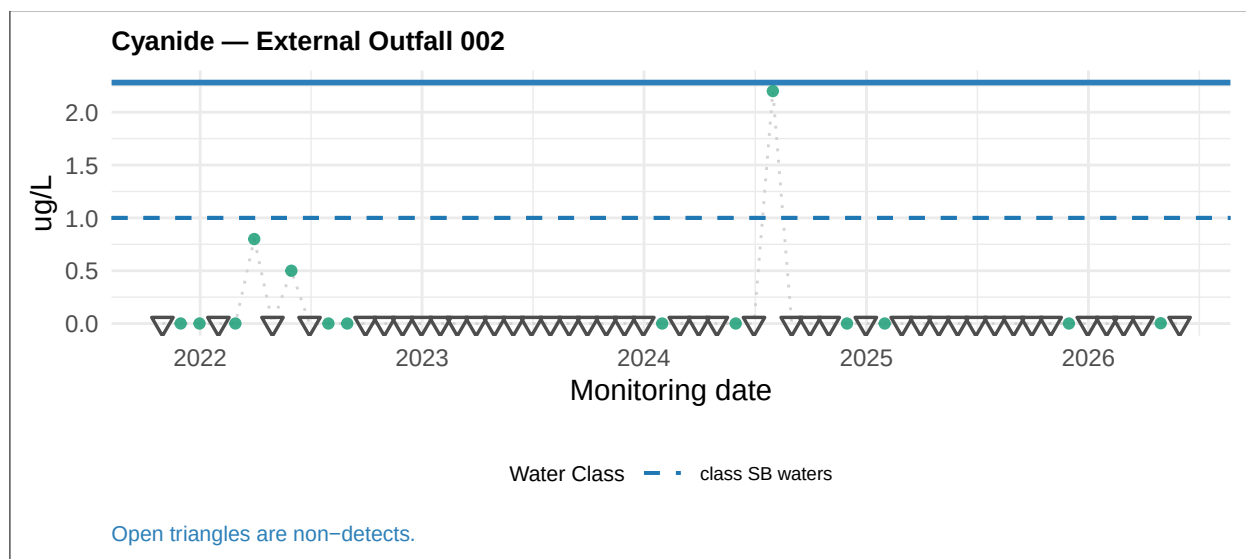
$$RWC = \frac{\max(x) \times MF}{DR} = \frac{54 \times 1}{1} = 55$$



4.3 Cyanide

4.3.1 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79232	class SB waters	56	No	1	2.28	YES



RWC Calculation

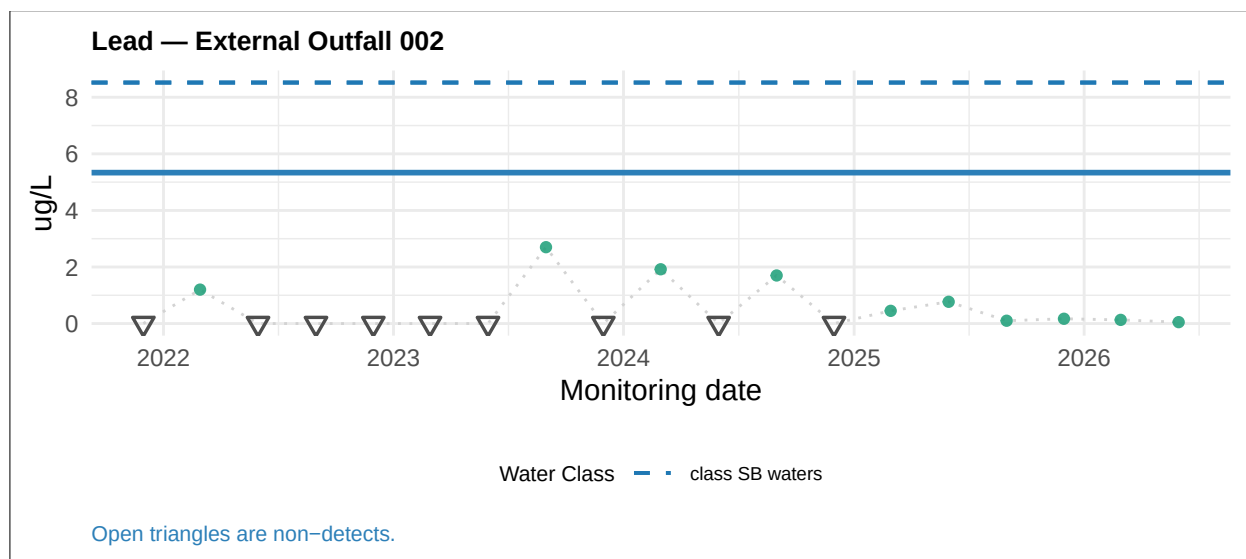
$$\begin{aligned}\sigma_g &= \sqrt{\ln(5^2 + 1)} = 1.8 \\ p &= (1 - 0.95)^{1/56} = 0.95 \\ z_p &= 1.6, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - 1.6) \times 1.8) = 1 \\ \text{RWC} &= \frac{\max(x) \times MF}{DR} = \frac{2.2 \times 1}{1} = 2.3\end{aligned}$$



4.4 Lead

4.4.1 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79252	class SB waters	19	No	8.52	5.34	NO



RWC Calculation

$$\sigma_g = \sqrt{\ln(1.7^2 + 1)} = 1.2$$

$$p = (1 - 0.95)^{1/19} = 0.85$$

$$z_p = 1.1, z_{\text{target}} = 1.6$$

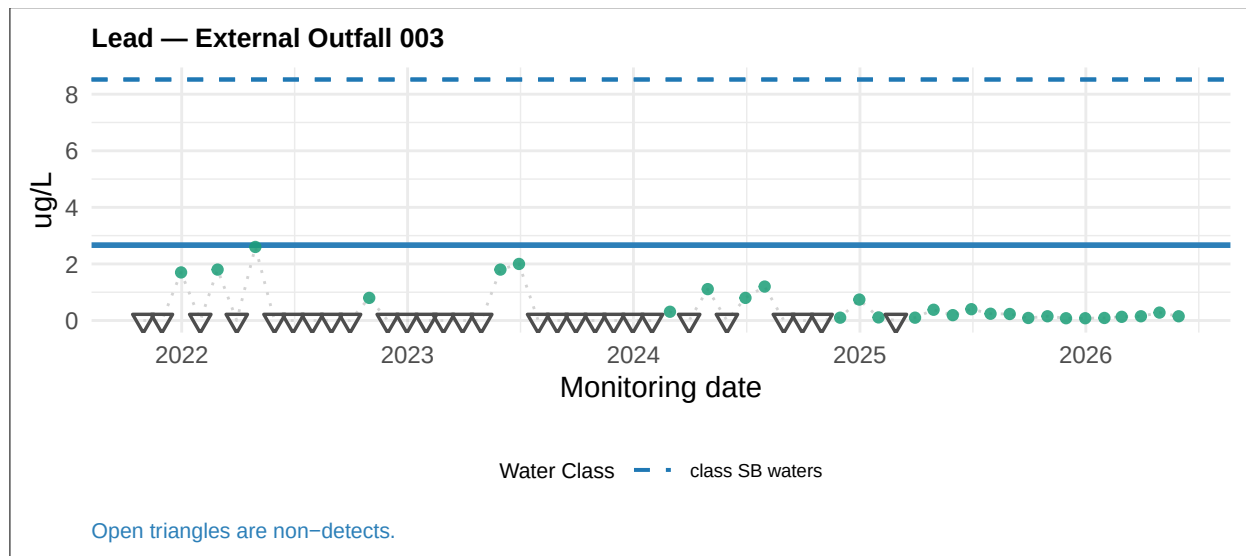
$$MF = \exp((1.6 - 1.1) \times 1.2) = 2$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{2.7 \times 2}{1} = 5.3$$



4.4.2 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79252	class SB waters	56	No	8.52	2.66	NO



RWC Calculation

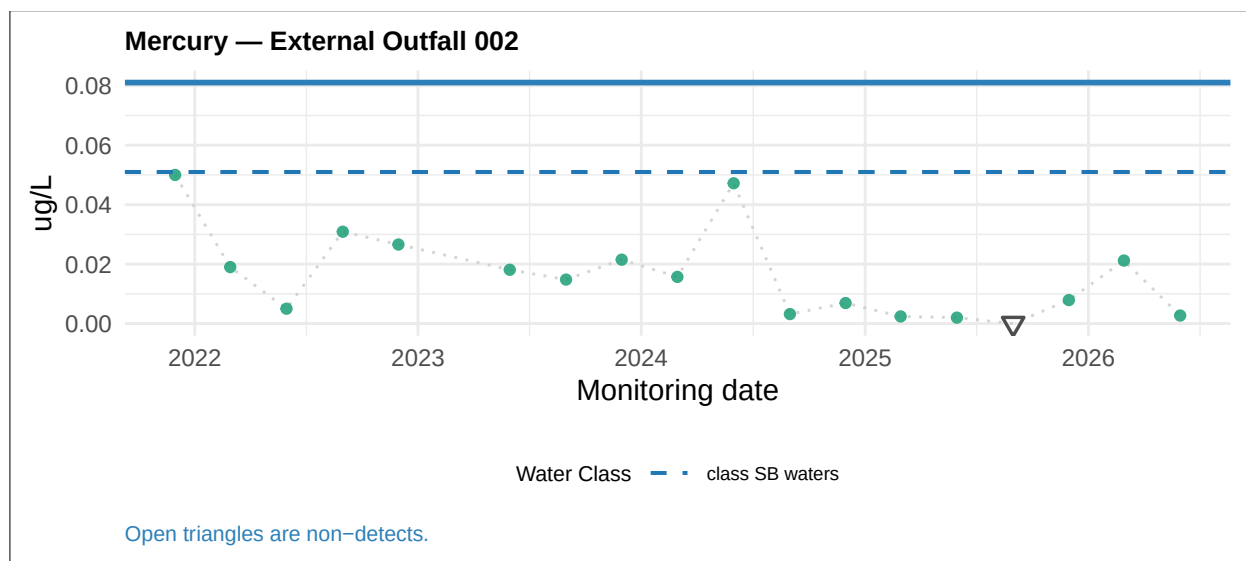
$$\begin{aligned}\sigma_g &= \sqrt{\ln(1.9^2 + 1)} = 1.2 \\ p &= (1 - 0.95)^{1/56} = 0.95 \\ z_p &= 1.6, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - 1.6) \times 1.2) = 1 \\ RWC &= \frac{\max(x) \times MF}{DR} = \frac{2.6 \times 1}{1} = 2.7\end{aligned}$$



4.5 Mercury

4.5.1 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79244	class SB waters	18	No	0.051	0.0811	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.91^2 + 1)} = 0.78$$

$$p = (1 - 0.95)^{1/18} = 0.85$$

$$z_p = 1, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1) \times 0.78) = 1.6$$

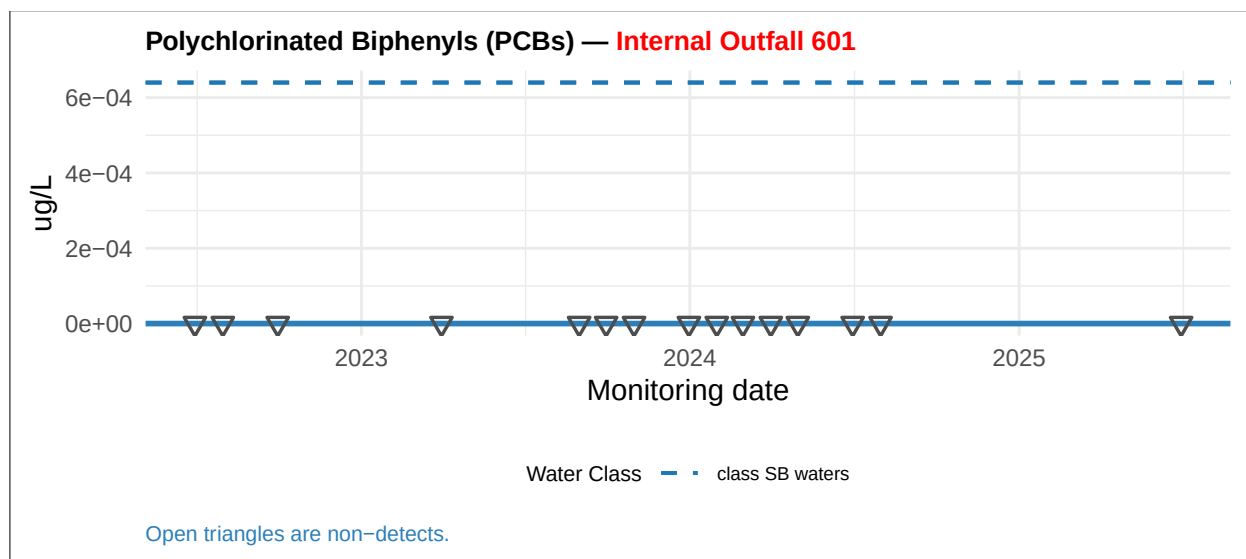
$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0.05 \times 1.6}{1} = 0.081$$



4.6 Polychlorinated Biphenyls (PCBs)

4.6.1 Internal Outfall 601

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79401	class SB waters	15	No	0.00064	0	NO



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/15} = 0.82$$

$$z_p = 0.91, z_{\text{target}} = 1.6$$

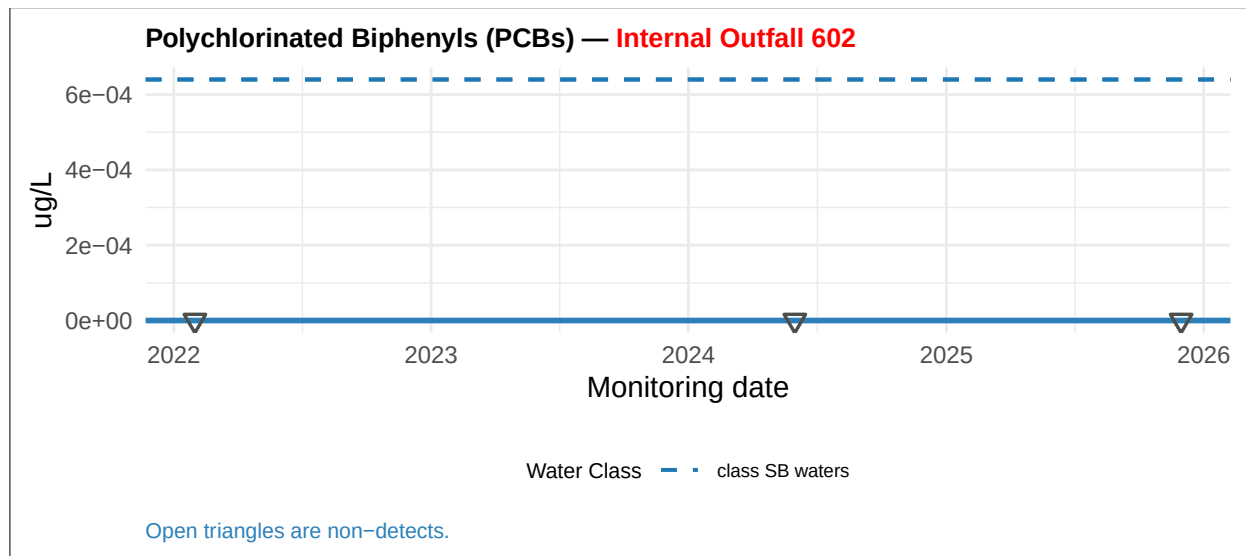
$$MF = \exp((1.6 - 0.91) \times 0.55) = 1.5$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0 \times 1.5}{1} = 0$$



4.6.2 Internal Outfall 602

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79401	class SB waters	3	No	0.00064	0	NO



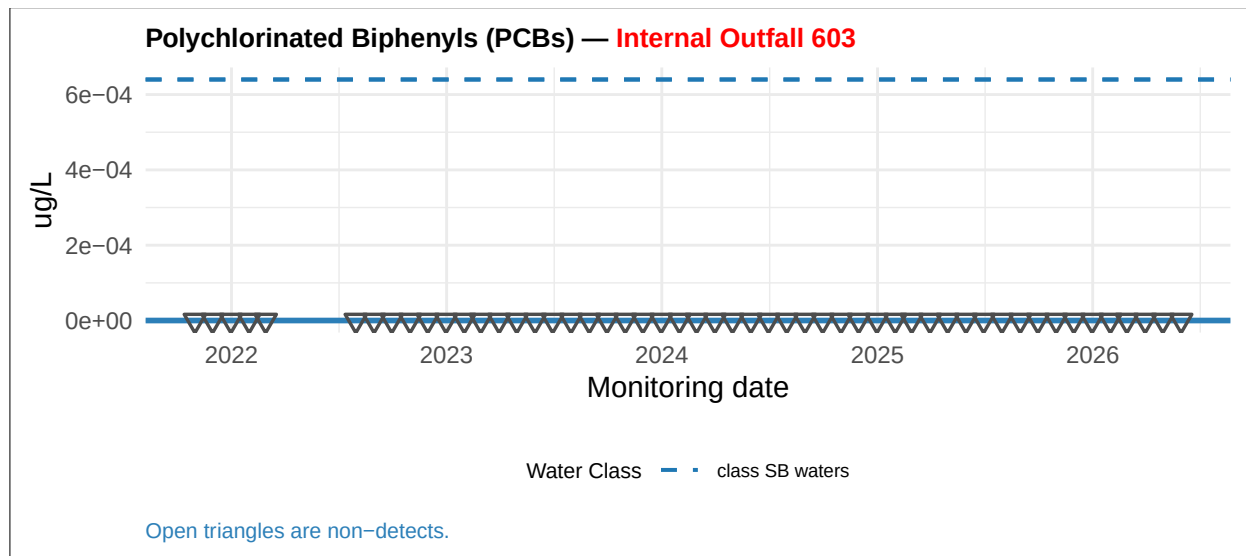
RWC Calculation

$$\begin{aligned}\sigma_g &= \sqrt{\ln(0.6^2 + 1)} = 0.55 \\ p &= (1 - 0.95)^{1/3} = 0.37 \\ z_p &= -0.34, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - -0.34) \times 0.55) = 3 \\ RWC &= \frac{\max(x) \times MF}{DR} = \frac{0 \times 3}{1} = 0\end{aligned}$$



4.6.3 Internal Outfall 603

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79401	class SB waters	52	No	0.00064	0	NO



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/52} = 0.94$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

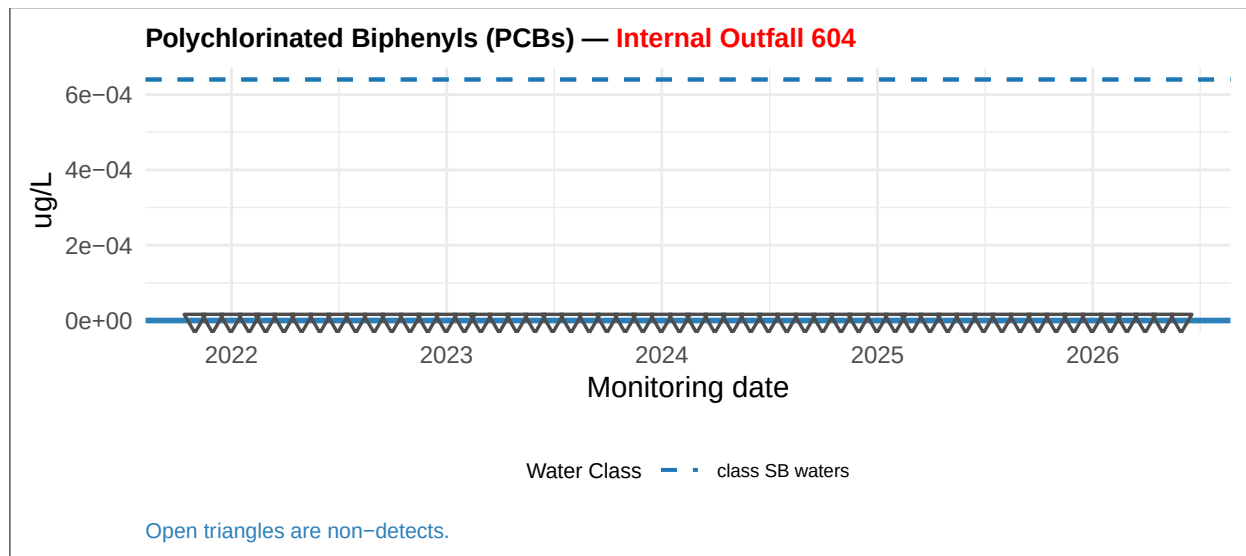
$$MF = \exp((1.6 - 1.6) \times 0.55) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0 \times 1}{1} = 0$$



4.6.4 Internal Outfall 604

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79401	class SB waters	56	No	0.00064	0	NO



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

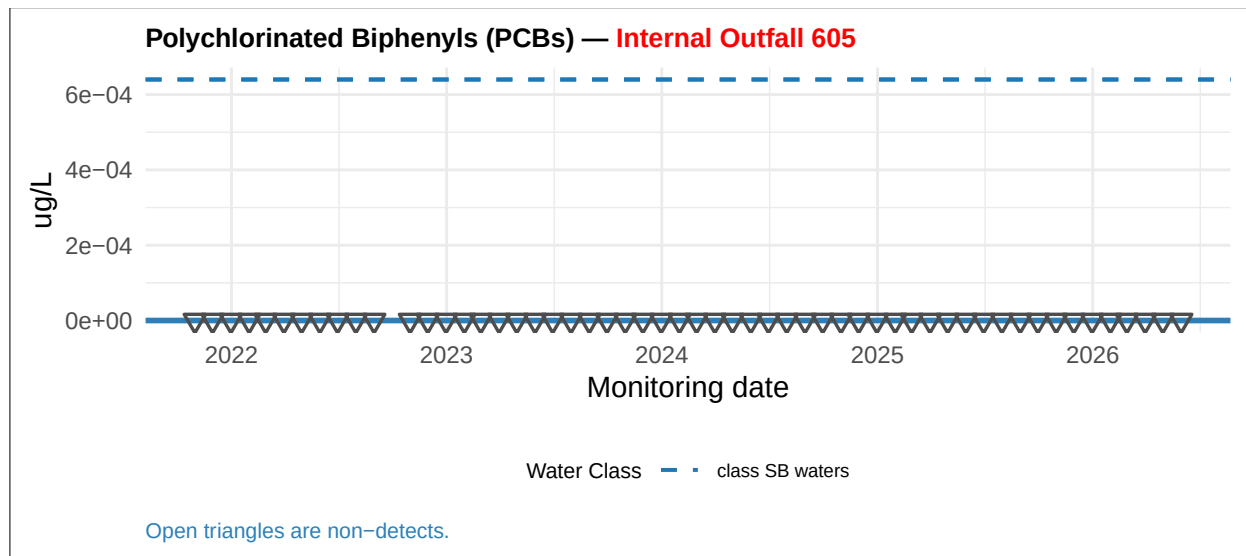
$$MF = \exp((1.6 - 1.6) \times 0.55) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0 \times 1}{1} = 0$$



4.6.5 Internal Outfall 605

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79401	class SB waters	55	No	0.00064	0	NO



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/55} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.6) \times 0.55) = 1$$

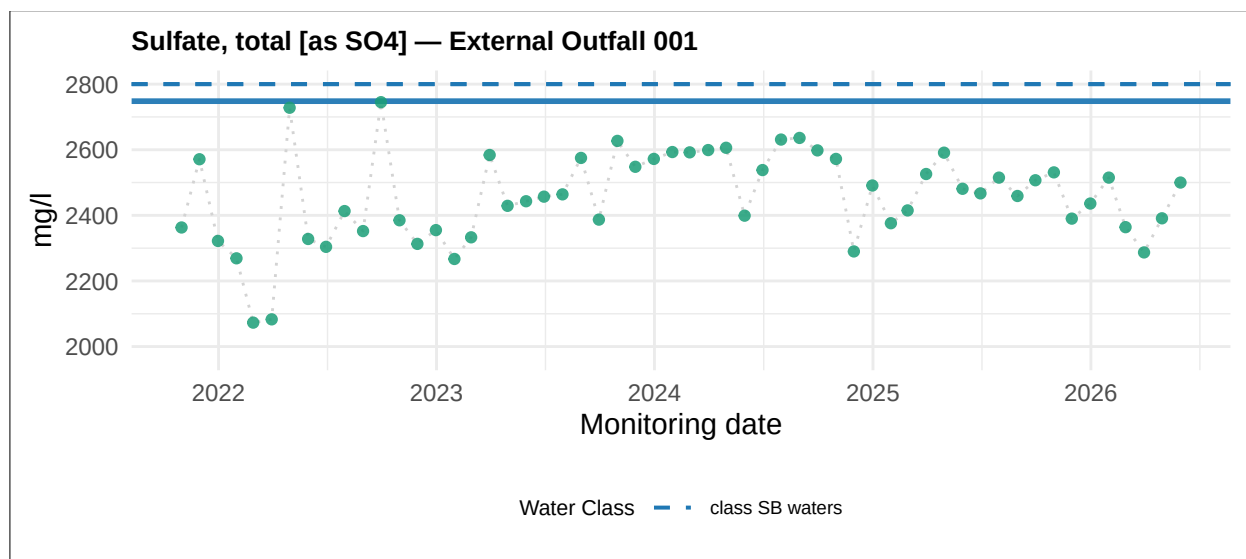
$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0 \times 1}{1} = 0$$



4.7 Sulfate, total [as SO₄]

4.7.1 External Outfall 001

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79597	class SB waters	56	No	2800	2750	NO



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.056^2 + 1)} = 0.056$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.6) \times 0.056) = 1$$

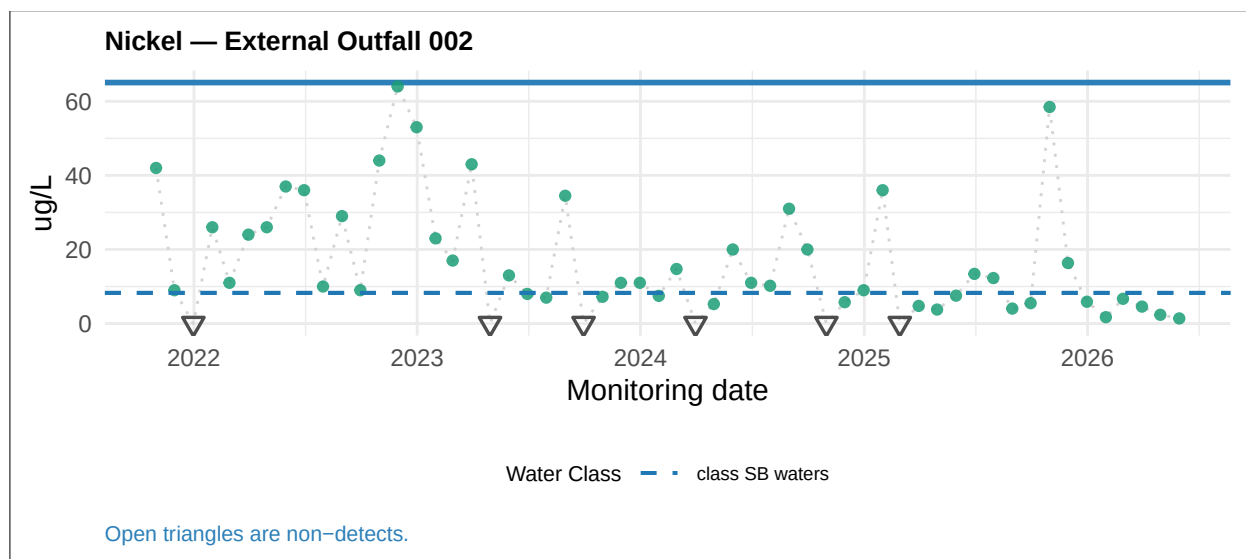
$$RWC = \frac{\max(x) \times MF}{DR} = \frac{2700 \times 1}{1} = 2700$$



4.8 Nickel

4.8.1 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79247	class SB waters	56	No	8.28	65.1	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.97^2 + 1)} = 0.82$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

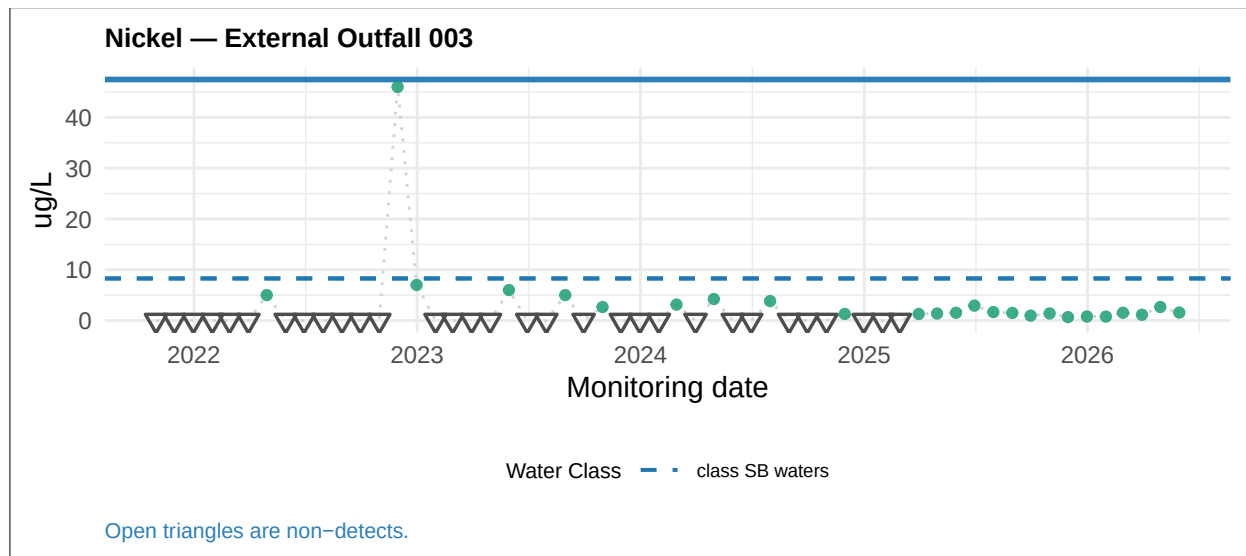
$$MF = \exp((1.6 - 1.6) \times 0.82) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{64 \times 1}{1} = 65$$



4.8.2 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79247	class SB waters	56	No	8.28	47.5	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(3.3^2 + 1)} = 1.6$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.6) \times 1.6) = 1$$

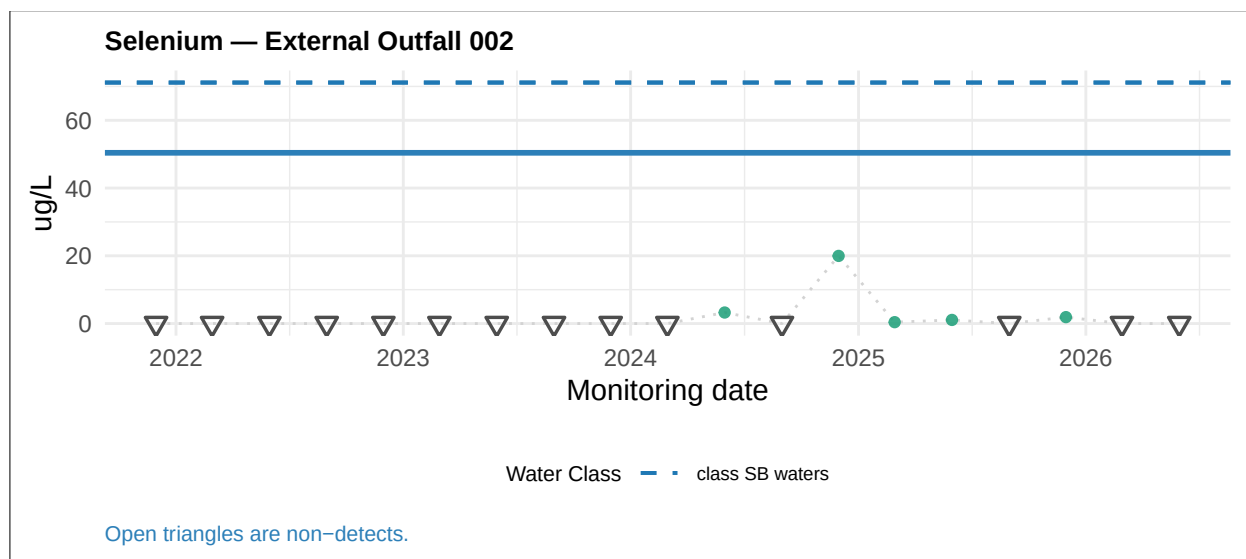
$$RWC = \frac{\max(x) \times MF}{DR} = \frac{46 \times 1}{1} = 47$$



4.9 Selenium

4.9.1 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79255	class SB waters	19	No	71.14	50.4	NO



RWC Calculation

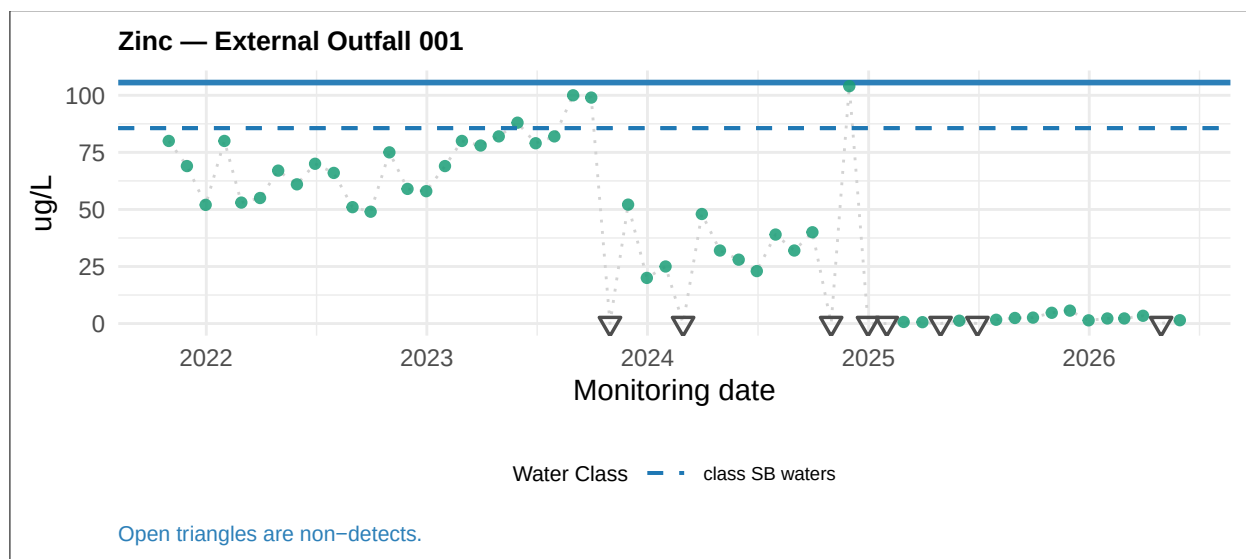
$$\begin{aligned}\sigma_g &= \sqrt{\ln(3.3^2 + 1)} = 1.6 \\ p &= (1 - 0.95)^{1/19} = 0.85 \\ z_p &= 1.1, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - 1.1) \times 1.6) = 2.5 \\ \text{RWC} &= \frac{\max(x) \times MF}{DR} = \frac{20 \times 2.5}{1} = 50\end{aligned}$$



4.10 Zinc

4.10.1 External Outfall 001

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79263	class SB waters	56	No	85.62	106	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.88^2 + 1)} = 0.76$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

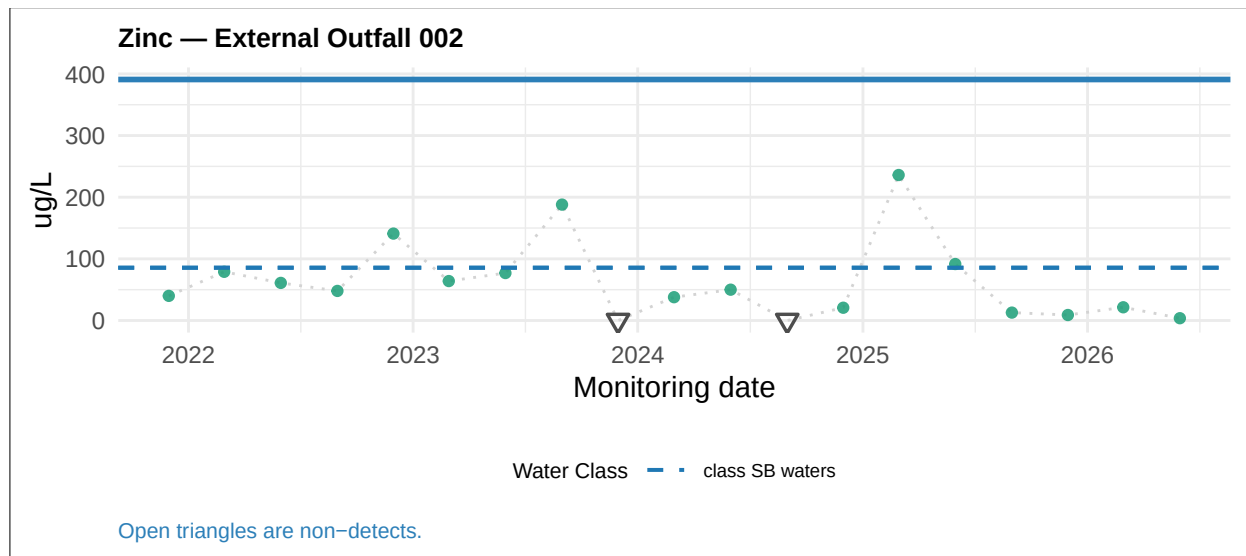
$$MF = \exp((1.6 - 1.6) \times 0.76) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{100 \times 1}{1} = 110$$



4.10.2 External Outfall 002

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79263	class SB waters	19	No	85.62	391	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(1^2 + 1)} = 0.85$$

$$p = (1 - 0.95)^{1/19} = 0.85$$

$$z_p = 1.1, z_{\text{target}} = 1.6$$

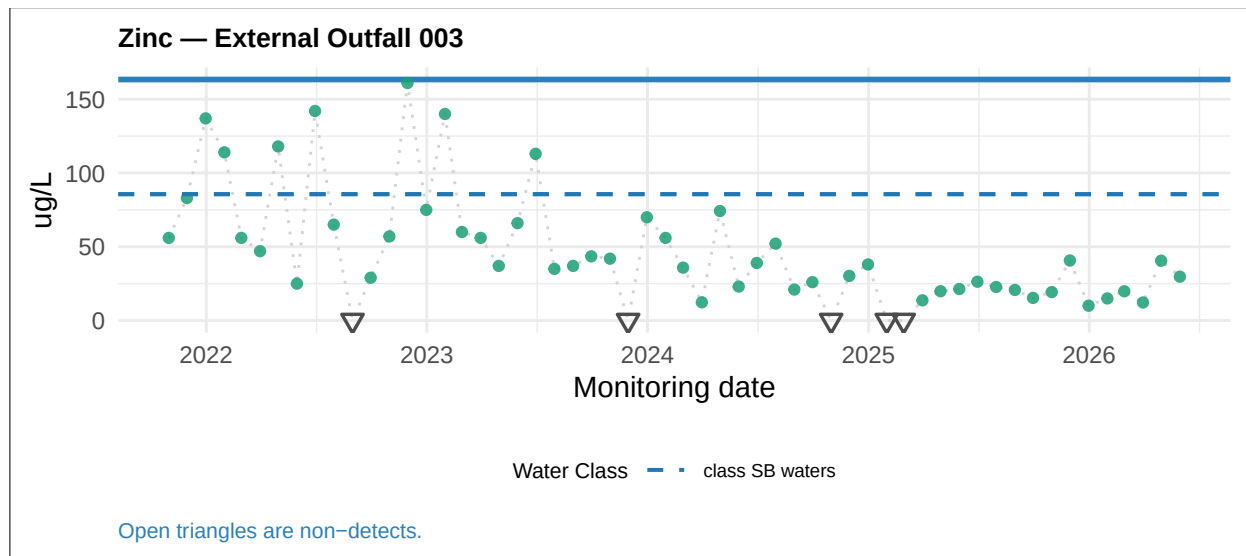
$$MF = \exp((1.6 - 1.1) \times 0.85) = 1.7$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{240 \times 1.7}{1} = 390$$



4.10.3 External Outfall 003

WQS ID	Water Class	Samples	Units Converted	WQS Criterion	RWC	RP
79263	class SB waters	56	No	85.62	163	YES



RWC Calculation

$$\sigma_g = \sqrt{\ln(0.84^2 + 1)} = 0.73$$

$$p = (1 - 0.95)^{1/56} = 0.95$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.6) \times 0.73) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{160 \times 1}{1} = 160$$



5 Parameters from Applications without DMR Data

Pollutant	Ref Param	# Samples
1,1,1-trichloroethane	34506	0
1,1,2,2-tetrachloroethane	34516	0
1,1,2-trichloroethane	34511	0
1,1-dichloroethylene	34501	0
1,2,4-trichlorobenzene	34551	0
1,2-dichlorobenzene	34536	0
1,2-dichloroethane	32103	0
1,2-dichloropropane	34541	0
1,2-diphenylhydrazine	34346	0
1,2-trans-dichloroethylene	34546	0
1,2-trans-dichloroethylene	85810	0
1,3-dichlorobenzene	34566	0
1,3-dichloropropylene	77163	0
1,4-dichlorobenzene	34571	0
2,4-dichlorophenol	34601	0
2,4-dimethylphenol	34606	0
2,4-dinitrophenol	34616	0
2,4-dinitrotoluene	34611	0
2-chloronaphthalene	34581	0
2-chlorophenol	34586	0
3,3-dichlorobenzidine	34631	0
3,4-benzofluoranthene	34230	0
4,4'-DDD	39310	0
4,4'-DDE	39320	0
4,4'-DDT	39300	0
4,6-dinitro-o-cresol	03615	0
Acenaphthene	34205	0
Acrolein	34210	0
Acrylonitrile	34215	0
Aldrin	39330	0
Anthracene	34220	0
Antimony	01095	0
Antimony	01097	0
Antimony	01268	0
Arsenic	01002	0
Benzene	34030	0
Benzidine	39120	0
Benzo (a) anthracene	34526	0
Benzo (a) pyrene	34247	0
Benzo (k) fluoranthene	34242	0
Bis (2-chloroethyl) ether	34273	0
Bis (2-ethylhexyl) phthalate	39100	0
Bromoform	32104	0
Butyl benzyl phthalate	34292	0
Cadmium	01027	0
Carbon tetrachloride	32102	0

*(continued)*

Pollutant	Ref Param	# Samples
Chlordane	51032	0
Chlorobenzene	34301	0
Chlorodibromomethane	34306	0
Chloroform	32106	0
Chrysene	34320	0
Di-n-butyl phthalate	51011	0
Dibenzo (a,h) anthracene	34556	0
Dichlorobromomethane	32101	0
Dieldrin	39380	0
Diethyl phthalate	34336	0
Dimethyl phthalate	34341	0
Dimethyl phthalate	34342	0
Endosulfan sulfate	34351	0
Endrin	39390	0
Endrin aldehyde	34366	0
Ethylbenzene	34371	0
Ethylbenzene	37371	0
Ethylbenzene	78113	0
Fluoranthene	34376	0
Fluorene	34381	0
Heptachlor	39410	0
Heptachlor epoxide	39420	0
Hexachlorobenzene	39700	0
Hexachlorobutadiene	34391	0
Hexachlorobutadiene	39702	0
Hexachlorocyclopentadiene	34386	0
Hexachloroethane	34396	0
Indeno (1,2,3-cd) pyrene	34403	0
Isophorone	34408	0
Methyl bromide	34413	0
Methylene chloride	34423	0
N-nitrosodi-n-propylamine	34428	0
N-nitrosodimethylamine	34438	0
N-nitrosodiphenylamine	34433	0
Nitrobenzene	34447	0
Pentachlorophenol	39032	0
Phenol	34694	0
Pyrene	34469	0
Silver	01077	0
Surfactants	47021	0
Surfactants	38260	0
Tetrachloroethylene	34475	0
Tetrachloroethylene	85814	0
Thallium	01059	0
Toluene	34010	0
Toluene	34481	0
Total nitrogen (as N)	00600	0
Total phosphorus	00665	0

*(continued)*

Pollutant	Ref Param	# Samples
Toxaphene	39400	0
Trichloroethylene	34485	0
Trichloroethylene	39180	0
Vinyl chloride	39175	0
alpha-BHC	39336	0
alpha-BHC	39337	0
beta-BHC	34255	0
beta-BHC	39338	0
beta-BHC	51007	0
gamma-BHC	39782	0
p-chloro-m-cresol	70012	0



6 Excluded Parameters

The following parameters had DMR data available but were excluded from the RP analysis at the direction of the user.

Parameter	Code	Flag Reason	Reason for Exclusion
Solids, settleable	00545	No crosswalk entry	Testing
Temperature, length of excursion	45600	No crosswalk entry	Testing
Flow, in conduit or thru treatment plant	50050	No crosswalk entry	Testing
Flow, Excluding Stormwater Discharge	5005E	No crosswalk entry	Testing
Solids, total suspended	00530	No crosswalk entry	Testing



7 Manual WQS Overrides

The following parameters had no matching WQS criterion for the selected water class and were evaluated using manually entered criteria. These criteria were supplied by the permit writer and are documented here as part of the permanent record. All manually entered criteria are flagged with a dagger symbol in the Coverage Summary table.

Parameter	Code	Flag Reason	Water Class(es)	Criterion Value	Unit	Crite- rion Type	Basis / Notes
Temperature, water deg. centigrade	00010	Wrong water class	class SB waters	30	deg C	NA	Testing